

Part a Verification of FastAPI - Pass the data to the API (Dynamic Routing)

Modify [main.py](#) file:

```
main.py 1, M X
datascientest-data-engineering > fast-api >
1  from fastapi import FastAPI
2
3
4  api = FastAPI(
5      title = "My API"
6  )
7
8  @api.get("/")
9  def get_index():
10     return {
11         "data": "Hello World"
12     }
13
```

Then run command **uvicorn main:api --reload**:

```
ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/fast-api/
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ uvicorn main:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [2559] using StatReload
INFO: Started server process [2561]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Open URL <http://127.0.0.1:8000/>

```
{
  "data": "Hello World"
}
```

Raw Parsed

Modify [main.py](#)

```
main.py 1, M X
datascientest-data-engineering > fast-api
1  from fastapi import FastAPI
2
3
4  api = FastAPI(
5      title = "My API"
6  )
7
8  @api.get("/")
9  def get_index():
10     return {
11         "data": "Hello World"
12     }
13
14  @api.get("/item/{itemid}")
15  def get_item():
16     return {
17         "route": "dynamic"
18     }
19
```

Open new terminal and run commands below:

```
ubuntu@ip-172-31-18-113:~$ cd datascientest-data-engineering/fast-api/
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/item/1
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 10:58:17 GMT
server: uvicorn
content-length: 19
content-type: application/json

{"route": "dynamic"}
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/item/my_item
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 10:58:38 GMT
server: uvicorn
content-length: 19
content-type: application/json

{"route": "dynamic"}
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Again modify [main.py](#):

```
main.py 1, M X
datascientest-data-engineering > fast-api >
1  from fastapi import FastAPI
2
3
4  api = FastAPI(
5      title = "My API"
6  )
7
8  @api.get("/")
9  def get_index():
10     return {
11         "data": "Hello World"
12     }
13
14  @api.get("/item/{itemid}")
15  def get_item(itemid):
16     return {
17         "route": "dynamic",
18         "itemid": itemid
19     }
20
```

Refresh URL http://127.0.0.1:8000/item/my_item

```
{
  "route": "dynamic",
  "itemid": "my_item"
}
```

Update [main.py](#) to the following:

```
main.py 1, M x
datascientest-data-engineering > fast-api > main.py > ...
1  from fastapi import FastAPI    Import "fastapi" co
2
3
4  api = FastAPI(
5      title = "My API"
6  )
7
8  @api.get("/")
9  def get_index():
10     return {
11         "data": "Hello World"
12     }
13
14  @api.get("/item/{itemid}")
15  def get_item(itemid):
16     return {
17         "route": "dynamic",
18         "itemid": itemid
19     }
20
21  @api.get("/item/{itemid}/description/{language}")
22  def get_item_language(itemid, language):
23     if language == "en":
24         return {
25             "itemid": itemid,
26             "description": "an object",
27             "language": "en"
28         }
29     else:
30         return {
31             "itemid": itemid,
32             "description": "u objet",
33             "language": "fr"
34         }
35
```

Open URL <http://127.0.0.1:8000/item/1/description/en>

```
{
  "itemid": "1",
  "description": "an object",
  "language": "en"
}
```

Raw Parsed

Modify the [main.py](#) file again:

```
main.py 1, M X
datascientest-data-engineering > fast-api > main.py > ...
1  from fastapi import FastAPI    Import "fastapi" c
2
3
4  api = FastAPI(
5      title = "My API"
6  )
7
8  @api.get("/")
9  def get_index():
10     return {
11         "data": "Hello World"
12     }
13
14  @api.get("/item/{itemid:int}")
15  def get_item(itemid):
16     return {
17         "route": "dynamic",
18         "itemid": itemid
19     }
20
21  @api.get("/item/{itemid}/description/{language}")
22  def get_item_language(itemid, language):
23     if language == "en":
24         return {
25             "itemid": itemid,
26             "description": "an object",
27             "language": "en"
28         }
29     else:
30         return {
31             "itemid": itemid,
32             "description": "u objet",
33             "language": "fr"
34         }
35
```

Run following commands in the terminal:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/item/1234
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 11:13:05 GMT
server: uvicorn
content-length: 33
content-type: application/json

{"route": "dynamic", "itemid": 1234}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/item/my_item
HTTP/1.1 404 Not Found
date: Thu, 12 Jun 2025 11:13:14 GMT
server: uvicorn
content-length: 22
content-type: application/json

{"detail": "Not Found"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Open URL <http://127.0.0.1:8000/item/1234>

```
{
  "route": "dynamic",
  "itemid": 1234
}
```

Open URL http://127.0.0.1:8000/item/my_item

```
{
  "detail": "Not Found"
}
```

Modify [main.py](#) file again:

```
main.py 1, M ×
atascientest-data-engineering > fast-api > main.py > ...
1  from fastapi import FastAPI    Import "fastapi" co
2
3
4  api = FastAPI(
5      title = "My API"
6  )
7
8  @api.get("/")
9  def get_index():
10     return {
11         "data": "Hello World"
12     }
13
14  @api.get("/item/{itemid:int}")
15  def get_item(itemid):
16     return {
17         "route": "dynamic",
18         "itemid": itemid
19     }
20
21  @api.get("/item/{itemid}/description/{language}")
22  def get_item_language(itemid, language):
23     if language == "en":
24         return {
25             "itemid": itemid,
26             "description": "an object",
27             "language": "en"
28         }
29     else:
30         return {
31             "itemid": itemid,
32             "description": "u objet",
33             "language": "fr"
34         }
35
36  @api.get("/item/{itemid:float}")
37  def get_item_float(itemid):
38     return {
39         "route": "dynamic",
40         "itemid": itemid,
41         "source": "float"
42     }
43
44  @api.get("/item/{itemid}")
45  def get_item_default(itemid):
46     return {
47         "route": "dynamic",
48         "itemid": itemid,
49         "source": "string"
50     }
51
```

Run following commands in terminal:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/item/1234
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 11:25:07 GMT
server: uvicorn
content-length: 33
content-type: application/json

{"route": "dynamic", "itemid": 1234}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/item/my_item
HTTP/1.1 200 OK
date: Thu, 12 Jun 2025 11:25:18 GMT
server: uvicorn
content-length: 56
content-type: application/json

{"route": "dynamic", "itemid": "my_item", "source": "string"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ curl -X GET -i http://127.0.0.1:8000/item/1.234
Command 'url' not found, did you mean:

command 'urh' from snap urh (2.9.3)
command 'curl' from snap curl (8.14.1)
command 'erl' from snap erlang (25.3)
command 'surl' from snap surl (0.8.0)
command 'yurl' from snap yurl (v0.6.3)
command 'curl' from deb curl (7.68.0-1ubuntu2.25)
command 'ur' from deb libur-perl (0.470+ds-1)
command 'ul' from deb bsdmainutils (11.1.2ubuntu3)
command 'uil' from deb uil (2.3.8-2build1)
command 'zurl' from deb zurl (1.11.0-1)
command 'erl' from deb erlang-base (1:22.2.7+dfsg-1ubuntu0.3)
command 'erl' from deb erlang-base-hipe (1:22.2.7+dfsg-1ubuntu0.3)

See 'snap info <snapname>' for additional versions.
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$
```

Open URL <http://127.0.0.1:8000/item/1234>

```
{
  "route": "dynamic",
  "itemid": 1234
}
```

Open URL http://127.0.0.1:8000/item/my_item

```
{
  "route": "dynamic",
  "itemid": "my_item",
  "source": "string"
}
```

Open URL <http://127.0.0.1:8000/item/1.234>

```
{
  "route": "dynamic",
  "itemid": 1.234,
  "source": "float"
}
```

Create a small API for user base [users.py](#). * N.B. Kill processes running for [main.py](#) file before running command **uvicorn users:api --reload**

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ lsof -i :8000
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
uvicorn 2559 ubuntu 3u IPv4 34431 0t0 TCP localhost:8000 (LISTEN)
python3 5188 ubuntu 3u IPv4 34431 0t0 TCP localhost:8000 (LISTEN)
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ kill -9 2559
[1]+  Killed                  uvicorn main:api --reload
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api$ uvicorn users:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reload process [5705] using StatReload
INFO: Started server process [5707]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Open URL <http://127.0.0.1:8000/users>

```
[
  {
    "user_id": 1,
    "name": "Alice",
    "subscription": "free tier"
  },
  {
    "user_id": 2,
    "name": "Bob",
    "subscription": "premium tier"
  },
  {
    "user_id": 3,
    "name": "Clementine",
    "subscription": "free tier"
  }
]
```

Open URL <http://127.0.0.1:8000/users/1>

```
{
  "user_id": 1,
  "name": "Alice",
  "subscription": "free tier"
}
```

Open URL <http://127.0.0.1:8000/users/2>

```
{
  "user_id": 2,
  "name": "Bob",
  "subscription": "premium tier"
}
```

Open URL <http://127.0.0.1:8000/users/3>

```
{
  "user_id": 3,
  "name": "Clementine",
  "subscription": "free tier"
}
```

Open URL <http://127.0.0.1:8000/users/1/name>

```
{
  "name": "Alice"
}
```

Open URL <http://127.0.0.1:8000/users/2/name>

```
{
  "name": "Bob"
}
```

Open URL <http://127.0.0.1:8000/users/3/name>

```
{
  "name": "Clementine"
}
```

Raw | Parsed

Open URL <http://127.0.0.1:8000/users/1/subscription>

```
{
  "subscription": "free tier"
}
```

Raw | Parsed

Open URL <http://127.0.0.1:8000/users/2/subscription>

```
{
  "subscription": "premium tier"
}
```

Raw | Parsed

Open URL <http://127.0.0.1:8000/users/3/subscription>

```
{
  "subscription": "free tier"
}
```

Raw | Parsed

Open URL <http://127.0.0.1:8000/docs>

default ^

GET / Get Index ^

GET /users Get Users ^

GET /users/{userid} Get User ^

GET /users/{userid}/name Get User Name ^

GET /users/{userid}/subscription Get User Subscription ^

Schemas ^

HTTPValidationError > Expand all object

ValidationError > Expand all object

Executed GET request to find **userid 1**:

GET / GET HTTP

GET /users Get Users

GET /users/{userid} Get User

GET /users/{userid}/name Get User Name

Parameters

Cancel

Name	Description
userid required	
any (path)	1

Execute Clear

Responses

Curl

curl -X 'GET' \ 'http://127.0.0.1:8000/users/1/name' \ -H 'accept: application/json'

Request URL

http://127.0.0.1:8000/users/1/name

Server response

Code	Details
200	Response body { "name": "Alice" } Download

Response headers

Opened URL <http://localhost:8000/redoc>

Search...

GET Get Index

GET Get Users

GET Get User

GET Get User Name

GET Get User Suscription

Get Index

Responses

> 200 Successful Response

Get Users

Responses

> 200 Successful Response

Get User

Responses

> 200 Successful Response

GET /

Response samples

200

Content type application/json

Copy

null

GET /users

Response samples

200

Content type application/json

Copy

null

GET /users/{userid}

Response samples

200

Content type application/json

Copy

null